

Practical # 4

Write the program to find the factorial of an inputted number.

```
1  #include <stdio.h>
2
3  int main() {
4      int num;
5      long long fact = 1;
6
7      printf("Enter a number: ");
8      scanf("%d", &num);
9
10     if(num < 0) {
11         printf("Factorial of a negative number doesn't exist.");
12     } else {
13         for(int i = 1; i <= num; i++) {
14             fact *= i;
15         }
16         printf("Factorial of %d is %lld", num, fact);
17     }
18
19     return 0;
20 }
```

```
F:\Practicals\lab4.exe
Enter a number: 4
Factorial of 4 is 24
-----
Process exited after 2.23 seconds with return value 0
Press any key to continue . . .
```

Algorithm:

1. Start
2. Input a number and store it in variable num.
3. If num < 0,
 Display "Factorial of negative number doesn't exist"
 Go to step 7.
4. Set fact = 1.
5. Set i = 1.
6. Repeat while $i \leq \text{num}$:
7. Multiply fact = fact * i
8. Increment $i = i + 1$
9. Display the value of fact.
10. End